9. Ideal gas and indistinguishability

Indistinguishability

- Let us review the arguments of key point 13 for weakly interacting particles.

- Identification of a microstate as a specification of the state of each and every particle, $i_1 \ldots i_N$ of particles $1 \ldots N$.
- Reorganisation of the partition function as

$$Z = \sum_{i_1 = i_2 \cdots = i_N} \exp(-\beta[\epsilon_{i_1} + \epsilon_{i_2} + \cdots + \epsilon_{i_N}]) = \left[\sum_{i_1} e^{-\beta \epsilon_{i_1}} \right] \cdots \left[\sum_{i_N} e^{-\beta \epsilon_{i_N}} \right] = [Z(1)]^N$$

- We must now recognise that these steps presuppose that the constituent particles are at least, in principle, **distinguishable** from one another, so that each may be thought of as carrying a personal identifying label.

Indistinguishability

- Specifically:

- The interpretation of 'microstate' presupposes that we can (in principle) point to each particle in turn and identify its state ('particle 33 is in state $i_{33} = 4$').

- The reformulated Z counts the contributions of many microstates which are in fact identical unless the particles are in some sense distinguishable. Thus for example a microstate with $i_1 = 1$ and $i_2 = 2$ is 'different' from a microstate with $i_1 = 2$ and $i_2 = 1$ only if we can distinguish between 'particle 1' and 'particle 2'.

Indistinguishability

- For model crystalline solids or magnets, there is nothing for us to worry about: the atoms in a solid are **localised** in space; their 'addresses' in space provide, in principle, the ingredients of a labelling scheme which renders them distinguishable.

- In the case of a gas, within a classical world view, there is also no problem. We may envisage 'christening' the atoms at some point and keeping track of their identities.

- But quantum physics tells us that particles are at best characterised not by trajectories, but by wave functions whose overlap makes it impossible to 'keep track of' which particle is which.
- We reach a profound conclusion:

- Key point 14: Identical non-localised particles must be treated as indistinguishable.

Accounting for indistinguishability

- Aim modify in a simple way the Boltzmann theory to take into account indistinguishability that arises from non-localisation.
- Roughly speaking (see Blundell-Blundell for full argument), the number of microstates of a system of non-localised particles is less by a factor of $1/N!$ compared to the case where the particles are treated distinguishably.
- To correct for the overcounting due to indistinguishability, we divide the partition function by $N!$

$$Z_{\text{indist.}} = \frac{1}{N!} \sum_{i_1, i_2 \cdots, i_N} \exp(-\beta[\epsilon_{i_1} + \epsilon_{i_2} + \cdots + \epsilon_{i_N}]) = \frac{1}{N!}[Z(1)]^N$$

where i_n is the state of particle n.

- This simple correction for non-localisation is only valid at low densities

Calculation of $Z(1)$

- Consider a single free particle in a cubic box of side L.
- Starting point: Schrödinger's equation for a free particle: $\left(-\frac{\hbar^2}{2M}\nabla^2 - \epsilon\right)\psi = 0$
- In $1d$: $\psi'' = -k^2\psi$ where $k^2 = \frac{2M\epsilon}{\hbar^2}$

  $\psi = A\sin kx$ with $k = \frac{n\pi}{L}$ and $n = 1,2,3,\ldots$

  Standing wave solutions.
  ψ vanishes at $x = 0,L$

- In $3d$: $\psi = A\sin k_x x \sin k_y y \sin k_z z$ with $k_x = \frac{n_x\pi}{L}$, $k_y = \frac{n_y\pi}{L}$, $k_z = \frac{n_z\pi}{L}$

$$\epsilon = \frac{\hbar^2}{2M}(k_x^2 + k_y^2 + k_z^2) = \frac{\hbar^2\pi^2}{2ML^2}(n_x^2 + n_y^2 + n_z^2)$$

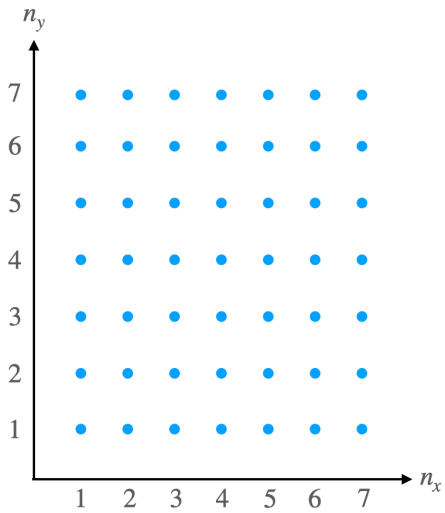
(See Chapter 11 of quantum mechanics notes)

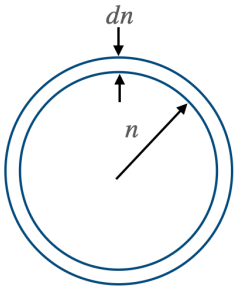
Calculation of $Z(1)$

- The task now is to sum over all allowed values of n_x, n_y, n_z:

$$Z(1) = \sum_{n_x,n_y,n_z} \exp(-\beta\epsilon(n_x, n_y, n_z)) = \exp(-\beta\epsilon(1,1,1)) + \exp(-\beta\epsilon(2,1,1)) + \ldots$$

- Energy only depends on sum of squares of quantum numbers n_x, n_y, n_z





- Approach: convert the sum into an integral in n-space over concentric shells of width dn

$$n = (n_x^2 + n_y^2 + n_z^2)^{1/2}$$

$$\sum_{n_x,n_y,n_z} [\cdots] \to \frac{1}{8}\int [\cdots] 4\pi n^2 dn$$

$$Z(1) \to \frac{\pi}{2}\int_0^\infty \exp\left(-\beta\frac{\hbar^2\pi^2 n^2}{2ML^2}\right) n^2 dn$$

- This is of course an approximation. It is a good approximation when $\dfrac{\hbar^2\pi^2}{2ML^2} \ll kT$